RPMGLOBAL

AMT EPF Adapter SDK

Assets 9.0

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Document History

Version	Date	Description	Author
1.0	13-Oct-2017	Initial Document	Andre Britz
1.1	7-Nov-2017	Updated with comments from Darryl Smith.	Andre Britz
1.2	18-Jan-2018	Various changes after reviewing XSD against this document.	Andre Britz
2.0.0	10-Jan-2019	Update Service References.	Andre Britz
2.0.1	9-Dec-2020	EPF 2.3/2.4 Support.	Grant Johnson
2.2.5	27-Jul-2022	Added RequiresSTO, ParentAsset and ConditionRating. Added reference to new configuration option ImportFuntionalLocationUseUidOnly.	Wayne Simpkins
2.2.6	24-Feb-2023	Added 'ClearExternalIdentifierForAllStrategyTasks' to configuration options.	Wayne Simpkins
2.3	19-Jul-2023	Miscellaneous grammar and spelling changes.	Wayne Simpkins
8.25	11-Sep-2023	Added details for ASYNC processing.	Wayne Simpkins
8.26	01-Nov-2023	Added chunk processing for ASYNC. Added new attributes PlanningGroupCode, WorkGroupCode and CatalogueProfileCode to ProcessAsset().	Wayne Simpkins
8.27	01-Jun-2024	Fixed incorrect detail about defaulting of DefaultCostCentreCode if blank in ProcessAsset().	Wayne Simpkins
8.28	01-Oct-2024	Added support for OAuth2 authentication. Added support for more functions in ASYNC.	Wayne Simpkins
8.28.2	10-Apr-2025	Added LastModifiedDateUTC and IncludeUsageCMCodes to GetAsset function.	Wayne Simpkins
9.0	17-Jul-2025	All inbound functions now support ASYNC processing. Moved details of synchronous/asynchronous processing, and authentication options to central overview document.	Wayne Simpkins

Version Amendments

The following modifications were made to this document since 1.1.

Section	Description
6.2	Field Name updates for: TermStartUsageMeasure >> TermStartUsageValue TermEndUsageMeasure >> TermEndUsageValue AnnualUsageMeasure >> AnnualUsageValue IndustryGroupName >> IndustryGroupCode
7	Warning added for FunctionalLocation business rules updates
8.1.1	CommissionDate business rule modified
8.4.2	Rule added to deal with the processing of ParentFunctionalLocationID
8.1.1	Business rules modification for AssetUtilisationTypeCode
5.2	Updated service reference. Spelling error

1 Business Background

The list of Assets and associated properties can be managed in AMT, or an external system.

The functionality supported by this integration can be used when an external system is the master of Assets, and AMT accepts updates directly from that system. This integration is not intended to be a bi-directional synchronisation, as information modified in AMT could potentially be overwritten by an incoming interface.

The alternative scenario allows for the Assets managed in AMT to be extracted and used in an external system. This is done on a request-only basis, as it is not the intention that AMT broadcasts the asset master data to other systems.

2 Considerations and Assumptions

When AMT is the “slave” system and all additions or updates to Assets are in an external “master” system, care should be taken when updating assets in AMT, as information may be overwritten by this interface.

An Asset in AMT is uniquely defined in one of two ways. The method is an AMT configuration option and should be considered during the design phase.

Option 1

a. Model

b. Serial Number

c. Registration Counter

Option 2

The Serial Number alone is used to identify the asset.

Note that these two identification options only apply to the Asset Adapter. The other adapters identify assets by option 1 only.

3 Decision Points and Configuration Options

The following AMT configuration settings will affect the behaviour of these interfaces, so the implementer will need to know these settings.

AMT Configuration Settings	Description
EquipmentImportUseSerialNumberAsKey	This setting determines how AMT defines unique assets. Refer to section 6.1, Asset Identification.
DefaultAnnualUtilisation	If no annual utilisation is provided, the asset is created as a usage-based asset with this annual utilisation rate, using the system default UOM.
DefaultEquipmentEndDate	If no end date or end usage is provided, the asset is created with this end date.
ImportFuntionalLocationUseUidOnly	When this flag is turned on, the AMT Adapter will use FunctionalLocationId only as a unique identifier. If the functional location name is provided, but different from that stored in the AMT Database, it will be updated accordingly.
ClearExternalIdentifierForAllStrategyTasks	If this flag is turned on, when terminating an asset, the external identifier will be removed from all strategy tasks for the asset in the current projection. The external identifier will be copied into the strategy task description before being removed.

4 Prerequisites

The following AMT configuration items must be completed before using these integration endpoints:

- Model
- UOM
- Cost Bearer
- Customer
- Model
- Asset Location
- Asset Criticality
- Asset Category
- Cost Responsibility
- Region
- Division
- Industry
- Industry Group
- Cost Centre

5 Supported Integration Patterns

5.1 Supported Functions

Function	Description
ProcessAsset	Add a new asset to AMT or update an existing asset if it already exists.
CancelAsset	Update the Asset Terminated flag.
GetAsset	List all assets matching the given search criteria.
ProcessFunctionalLocation	Add a new Functional Location to AMT or update an existing one if it already exists.
CancelFunctionalLocation	Delete a functional location from AMT.
ProcessFunctionalLocationRelationship	Create or update the functional location hierarchy (the relationships between functional locations).
GetFunctionalLocation	List all functional locations matching the given search criteria.

5.2 Integration Service Address

More details regarding the general features of the integration service can be found in the Service Overview documentation. In order to use this service, the following endpoints are available for various authentication scenarios. These may not all be available, depending on the deployment configurations at a specific site.

Basic authentication	Domain, username and password provided when establishing the connection
https://<Server>:<port>/AssetsIntegrationService/AssetsIntegration/BasicHttpBinding_Https_Basic	
http://<Server>:<port>/AssetsIntegrationService/AssetsIntegration/BasicHttpBinding_Http_Basic	

Windows authentication	The Windows credentials of the current Windows account are used to establish the connection. This is the default method if no binding is specified in the URL.
https://<Server>:<port>/AssetsIntegrationService/AssetsIntegration/	
http://<Server>:<port>/AssetsIntegrationService/AssetsIntegration/	

OAUTH2 authentication	Client authentication is used to retrieve an OAUTH2 access token. The token is passed in the SOAP header when calling the AMT EPF Adapters.
https://<Server>:<port>/AssetsIntegrationService/AssetsIntegration/BasicHttpBinding_Https_Token	
http://<Server>:<port>/AssetsIntegrationService/AssetsIntegration/BasicHttpBinding_Http_Token	

<Server> is the fully qualified name of the server hosting the service.

Note that Windows authentication is being deprecated.

5.3 Synchronous Transaction Patterns

Synchronous integration should align with the transaction pattern as shown in Figure 1.

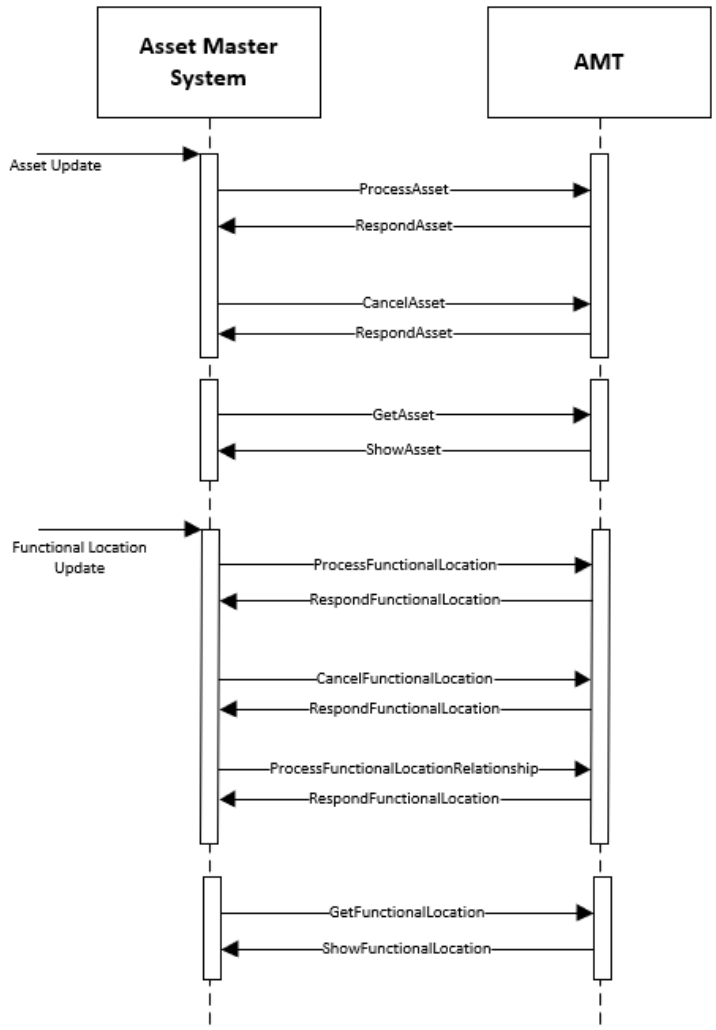


Figure 1 Synchronous Integration Patterns

6 Field Descriptions

The fields in the following tables are used for this integration. The data types for each field are described in the relevant XML schema definition.

6.1 Asset Identification

These fields are always used to identify assets in AMT. The option is dependent on the AMT configuration.

6.1.1 EquipmentImportUseSerialNumberAsKey=0

SerialNumber	Manufacturer's Serial Number for the Asset.
ModelCode	The Asset Model Code as configured in AMT.
AssetRegistrationCounter	Registration Counter of the asset, also called "Contract Enrolment Number".

6.1.2 EquipmentImportUseSerialNumberAsKey=1

In this case the model is still a required field but not used to evaluate uniqueness during processing.

Field Name	Description
SerialNumber	Manufacturer's Serial Number for the Asset.

6.2 Asset

Field Name	Description
AssetName	How the asset is to be named in AMT. This needs to be unique within AMT.
AssetDescription	Description of the asset.
ManufacturerName	The AMT configured manufacturer of the asset.
AssetReference	Reference text of the asset.
AssetNumber	The asset number as in the master system.
Fleet	The fleet the asset has been assigned to.
YearOfManufacture	The year the asset was manufactured.
CommissionDate	This is the original Commission Date of the asset.
DefaultUOMCode	While an asset in AMT can have multiple UOMs, this field defines the asset's default UOM.
TermStartDate	The date of the first use of the asset, or contract start date.
TermEndDate	The date of the last use of the asset, or contract completion.
UsageProfile	Defines the Usage Profile (Advanced Utilisation) attached to the asset.
Owner	Asset Business Owner.
TermStartUsageValue	The usage measurement on the term start date.

Field Name	Description
TermEndUsageValue	The usage measurement on the term end date.
AnnualUsageValue	Annual usage of the asset based on the UOM.
AssetPlanningTypeCode	The AMT configured planning type.
AssetUtilisationTypeCode	The AMT configured utilisation type.
AssetTerminatedIndicator	Shows if the asset is still in use or not.
LocationCode	Asset Location as configured in AMT.
RegionCode	Asset Region as configured in AMT.
DivisionCode	Asset Division as configured in AMT.
CriticalityCode	Asset Criticality as configured in AMT.
CategoryCode	Asset Cost Category as configured in AMT.
ResponsibilityCode	Asset Cost Responsibility as configured in AMT.
IndustryGroupCode	Asset Industry Group as configured in AMT.
CustomerCode	Asset Customer as configured in AMT.
AssignedToName	The person who has primary responsibility for the asset.
AssignedToPhone	Phone Number of the assigned to person.
AssignedToEmail	Email of the assigned to person.
AssignToAddress	Address of the assigned to person.
DefaultCostCentreCode	Asset Cost Centre Code as configured in AMT.
DefaultCostBearerCode	The default AMT Cost Bearer as configured in AMT.
BudgetPricingDate	Asset Budget Pricing Date.
BudgetMarginParts	Budget Margin for Parts for the asset.
BudgetMarginLabour	Budget Margin for Labour for the asset.
BudgetMarginMisc	Budget Margin for Misc for the asset.
AssetAutoCreateProjectionIndicator	Indicates whether or not to automatically create the Asset's Current Projection using the Centreline linked to the Model as the template.
RequiresSTO	Indicates if any performing any work for the asset requires an STO.
ParentAsset	Indicates the parent asset for this asset. Currently only applicable in ProcessAsset function. Not currently implemented for ShowAsset response.
ConditionRating	Condition rating for the Asset. Currently only applicable in ProcessAsset function. Not currently implemented for ShowAsset response.
PlanningGroupCode	Code for a planning group as configured in AMT. Only applicable to ProcessAsset().
WorkgroupCode	Code for a work group as configured in AMT. The work group must belong to the same site as the asset. Only applicable to ProcessAsset().
CatalogueProfileCode	Code for the planning catalogue as configured in AMT. Only applicable to ProcessAsset().

6.2.1 Usage and CM Codes

The Asset definition includes a list of CM and usage codes for the asset. These fields only apply to the ShowAsset message when responding to a GetAsset message request with IncludeUsageCMCodes set to 1.

The list of CM and usage codes does not apply to the ProcessAsset function.

The UsageCodeList provides a list of valid usage codes for the asset.

Field Name	Description
UsageCode	A valid AMT UOM code for the asset. This code can be used to call back into AMT using the ProcessUsageReading function of the Asset Statistic Adapter.
UsageDesc	Short description of the usage code.

The CMCodeList provides a list of valid CM measurement definitions for the asset. This provides codes that can be used to call back into AMT using the ProcessMeasurement function of the Asset Statistic Adapter.

Field Name	Description
CMCode	A valid AMT CM code for the asset.
CMDesc	Description of the condition monitoring code.
CMUOM	The unit of measure for the CM code.
CMMeasurementTypeCode	The measurement type for the CM code.
CMMeasuremenTypeDesc	Description of the condition monitoring measurement type.
CMCategory	The condition monitoring category.
CMCompartment	The relevant compartment of the machine for the CM code.

6.3 FunctionalLocation Identification

These fields must be provided in all cases and must be independently unique.

Field Name	Description
FunctionalLocationID	A unique identifier for the functional location.
FunctionalLocationName	The functional location name.

6.4 FunctionalLocation

Field Name	Description
FunctionalLocationDescription	Description of the Functional Location.
FunctionalLocationCategoryCode	The Functional Location Category Code as configured in AMT.
FunctionalLocationTypeCode	The Functional Location Type Code.
ParentFunctionalLocationID	The unique identifier for the Parent of this functional location.
ParentFunctionalLocationName	The Functional Location Name of the Parent of this functional location.
ExternalObjectType	The type of object in the external system.

Field Name	Description
ModelCode	The asset Model Code as configured in AMT.
SerialNumber	Manufacturer's Serial Number for the asset.
RegistrationCounter	Registration Counter of the asset, also called "Contract Enrolment Number".
FunctionalLocationTerminatedIndicator	Shows if Functional Location is in use or not.
FunctionalLocationAttribute	The name and value of the Functional Location Attribute, as configured in AMT.
RequiresSTO	Indicates if any performing any work for the Functional Location requires an STO. Not currently implemented for ShowFunctionalLocation response.

6.5 FunctionalLocation Relationship

Field Name	Description
ParentFunctionalLocationID	The unique identifier for the Parent of this functional location.
ParentFunctionalLocationName	The Functional Location Name of the Parent of this functional location.

7 Error Handling

The following valid errors are returned from this interface.

Error Condition	Description
The provided document has an unrecognised format. Various XML format errors are returned.	Check the format of the schema document against those provided. Errors can be related to well-formed XML issues, or failure to validate the document against the XSD.
The provided value for <ElementCode> does not have the correct datatype.	Check datatypes of elements, especially date formats. Note that date and datetime conventions are used.
Mandatory element <ElementName> not provided.	An expected xml element has not been provided.
Mandatory attribute <AttributeName> not provided.	An expected XML attribute has not been provided.
The provided code for <ElementCode> does not have a corresponding AMT configuration item.	AMT makes extensive use of pre-configured code lists. Relevant codes must be preconfigured prior to use of the interface.
A functional location update has been requested, but no corresponding asset has been found with: SerialNumber: <SerialNumber> ModelCode: <ModelCode> AssetRegistrationCounter:<RegistrationCounter>	This results when the external system tries to link a functional location for an asset that does not exist in AMT yet.

The following valid warnings are returned from this interface. These affected items will still be processed.

Warning Condition	Description
A functional location update has been requested, but no corresponding functional location has been found with: FunctionalLocationID: <FunctionalLocationID> FunctionalLocationName: <FunctionalLocationName>	This results when the external system tries to create a function location relationship, but functional location does not exist in AMT yet.

8 Integration Functions

Only fields with specific rules have been listed. All other fields should conform to the datatype rules as defined in the schema definition files, and are optional.

8.1 ProcessAsset

This interface is used to add new assets to AMT or edit existing assets.

8.1.1 Business Rules

Field Name	Business Rules	Optional/ Required
SerialNumber	This field is always used to uniquely identify a part within AMT. Depending on the AMT configuration, this field can be used together with ModelCode and AssetRegistrationCounter to uniquely identify the part within AMT.	R
ModelCode	The value provided should match a configured **Model** code within AMT. Depending on the AMT configuration, this field is used together with the Serial Number and AssetRegistrationCounter to uniquely identify the part within AMT.	R
AssetRegistrationCounter	Depending on the AMT configuration, this field is used together with the Serial Number and ModelCode to uniquely identify the part within AMT. If not provided, then the currently active registration is used.	O
AssetName	The asset name must be unique within AMT.	R
Fleet	The value provided should match a configured **Fleet** code within AMT.	O
CommissionDate	This must be the same or earlier than the TermStartDate. If no value is provided, the value of TermStartDate will be used.	O
TermStartDate	The date of first use of the asset, or contract start date.	R
CommissionDate	This must be the same or earlier than the TermStartDate. If no value is provided, the value of TermStartDate will be used.	O
AssetPlanningTypeCode	Only the following values are accepted: • **0** – No Planning • **1** – Advance Planning • **2** – Simple Planning If no value is provided, 0 is used.	O

Field Name	Business Rules	Optional/ Required
AssetUtilisationTypeCode	Only the following values are accepted: • **0** – No Utilisation • **1** – Simple Utilisation • **2** – Advanced Utilisation If no value is provided, 0 is used. Asset utilisation in AMT cannot be changed from "Advanced" to "Simple or No Utilisation" or from "Simple" to "No Utilisation". Therefore, if the asset's utilisation type passed by the interface is less than the existing utilisation type set against the equipment in AMT already, the UtilisationType passed by the interface will be ignored (no change will be made to the asset's utilisation type in AMT). If the UtilisationType passed by the interface is the same or greater than the asset's existing utilisation type, the UtilisationType will be updated based on the AssetUtilisationTypeCode value passed in the interface.	O
DefaultUOMCode	The value provided should match a configured **UnitOfMeasure** code within AMT. This field is only used for simple utilisation assets. Date-based assets do not use this value. Advanced utilisation assets do not use this value.	
UsageProfileName	This must be provided when AssetUtilisationTypeCode = 2. If the profile name does not exist within AMT, one will be created with the Default Calendar Year usage value.	
TermStartUsageMeasure	This field is ignored for Date-based assets.	
AnnualUsageMeasure	If no value is provided, the default value as configured in AMT *DefaultAnnualUtilisation.* This field is ignored for Date-based assets.	
TermEndDate	This must be after the start date. If no value is provided, the default end date is used, as configured in *DefaultEquipmentEndDate.*	
TermEndUsageMeasure	If no end usage is supplied, the end usage is calculated from the end date.	
AssetTerminatedIndicator	Only the following values are accepted: • **0** (Not Terminated) • **1** (Terminated) If no value is provided, 0 is used.	
LocationCode	The value provided should match a configured **Asset Location** code within AMT.	
CriticalityCode	The value provided should match a configured **Asset Criticality** code within AMT.	

Field Name	Business Rules	Optional/ Required
CategoryCode	The value provided should match a configured **Asset Category** code within AMT.	
ResponsibilityCode	The value provided should match a configured **Responsibility** code within AMT.	
RegionCode	The value provided should match a configured **Region** code within AMT.	
DivisionCode	The value provided should match a configured **Division** code within AMT.	
IndustryGroupCode	The value provided should match a configured **Industry Group** code within AMT.	
CustomerCode	The value provided should match a configured **Customer** code within AMT.	
DefaultCostCentreCode	The value provided should match a configured **Cost Centre** within AMT.	
DefaultCostBearerCode	If no value is provided, the default **Cost Bearer** is used.	
BudgetPricingDate	If no value is provided, **TermStartDate** is used. This is only used for the budget projection (if it exists).	
BudgetMarginParts	If no value is provided, a value of 0 is used. This is only used for the budget projection (if it exists).	
BudgetMarginLabour	If no value is provided, a value of 0 is used. his is only used for the budget projection (if it exists).	
BudgetMarginMisc	If no value is provided, a value of 0 is used. This is only used for the budget projection (if it exists).	
AssetAutoCreateProjectionIndicator	If set to true, a current projection is automatically created based on the centreline associated with the asset's model. For this to work, the following must also be true: • The model must have a centreline associated. • There must be no tasks in the current projection. • The asset must not be terminated. The asset must not be flagged to be excluded from the sales forecast. The territory indicator of the asset must be set to "O".	
RequiresSTO	Only the following values are accepted: • 0 – STO not required (false). • 1 – STO is required (true) If no value is provided, 0 is used.	O

Field Name	Business Rules	Optional/ Required
ParentAsset	Must match an existing AMT asset as determined by the combination of ParentAsset.ModelCode, ParentAsset.SerialNumber and ParentAsset.RegistrationCounter.	O
ConditionRating	The value provided should match a configured **Asset Condition** code within AMT.	O
PlanningGroupCode	Must match an existing AMT planning **Group Code**.	O
WorkGroupCode	Must match an existing AMT **Work Group Code**. The work group code must be valid on the site of the asset.	O
CatalogueProfileCode	Must match an existing AMT **Catalogue Profile Code**.	O

8.1.2 Processing Rules

1. One or more assets can be included in the same message.
2. Each asset is processed individually. Failure of a single asset load is reported, and processing continues.
3. The configuration setting **EquipmentImportUseSerialNumberAsKey** will determine what method is used to determine if assets exist.
4. If the asset does not exist:
    a. The asset is created as a new asset.
    b. Any value against a provided optional tag is used.
    c. If an optional tag is not provided, the value is not set in AMT unless a specific business rule exists to manage this case.
5. If the asset does exist:
    a. The asset will be updated with new provided values.
    b. Any value against a provided tag will be overwritten.
    c. If no tag is provided, the existing values will remain as-is.
6. Assets can be duplicated - the asset processed last will take precedence.
7. **RespondAsset** is returned upon completion of processing.

8.1.3 Data Structure

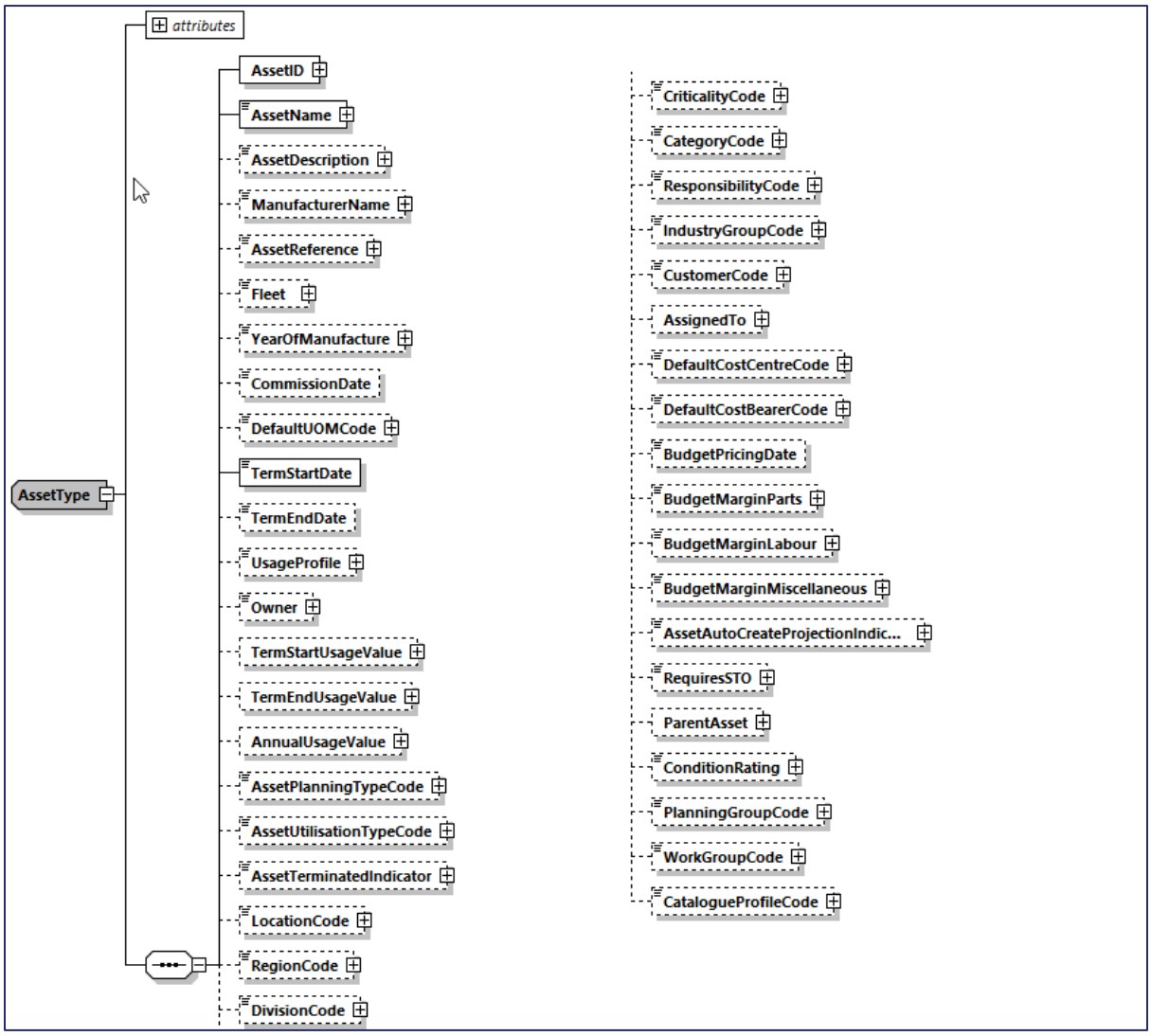


Figure 2 ProcessAsset Data Area

8.2 CancelAsset

This interface is used to remove assets from AMT. Rather than deleting the asset within AMT, the **Asset Terminated** flag is set to **True**.

8.2.1 Business Rules

Field Name	Business Rules	Optional/ Required
SerialNumber	This field is always used to uniquely identify a part within AMT. Depending on the AMT configuration, this field can be used together with ModelCode and AssetRegistrationCounter to uniquely identify the part within AMT.	R
ModelCode	Depending on the AMT configuration, this field is used together with Serial Number and AssetRegistrationCounter to uniquely identify the part within AMT.	R
AssetRegistrationCounter	Depending on the AMT configuration, this field is used together with the Serial Number and Model Code to uniquely identify the part within the AMT. If not provided, the currently active registration is used.	O

8.2.2 Processing Rules

1. One or more assets can be included in the same message.
2. Each asset is cancelled individually. Failure of a single asset is reported, and processing continues.
3. The configuration setting **EquipmentImportUseSerialNumberAsKey** will determine what method is used to determine if assets exist.
4. Assets can be duplicated within the message.
5. If the asset does exist, the **Asset Terminated** flag is set to **True**.
6. If the asset does not exist in AMT, no further action is taken.
7. **RespondAsset** is returned upon completion of the process.

8.2.3 Data Structure

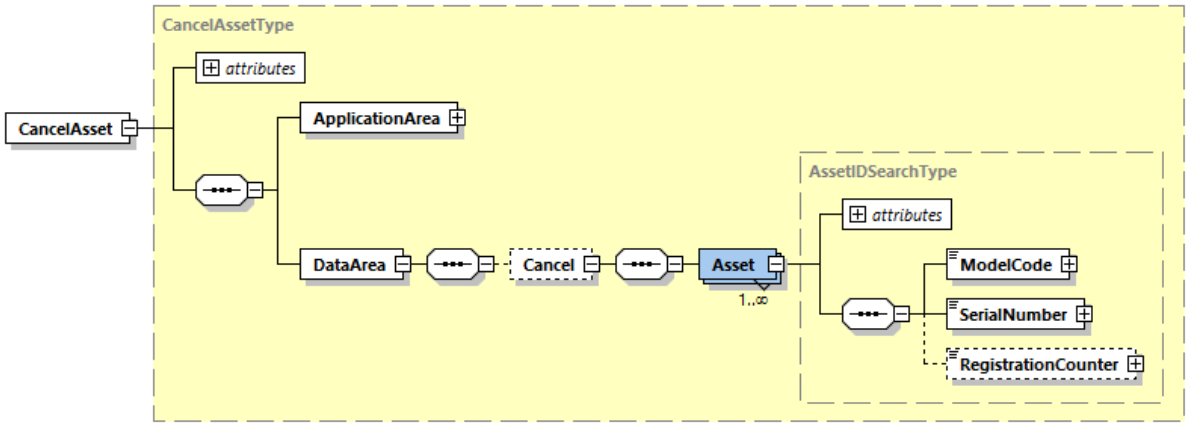


Figure 3 CancelAsset DataArea

8.3 GetAsset

This interface is used to retrieve a list of assets from AMT.

8.3.1 Business Rules

Field Name	Business Rules	Optional/ Required
Serial Number	The complete SerialNumber is matched.	O
ModelCode	The complete ModelCode is matched.	O
AssetName	The complete AssetName is matched.	O
LastModifiedDateUTC	Assets modified since this date are returned in the response message.	O
IncludeUsageCMCodes	Indicates whether to retrieve valid Usage codes and CM codes for assets returned in the response payload. Valid values are : • 0 – Do not return usage and CM codes • 1 – Return usage and CM codes.	O

8.3.2 Processing Rules

1. Any provided fields will be matched with existing Assets in AMT.
2. All items within a list element are considered as a logical OR

   Each list is considered as a logical AND
3. The use of a non-existent model code will result in an error condition.
4. **ShowAsset** is returned upon completion of the process.

8.3.3 Data Structure

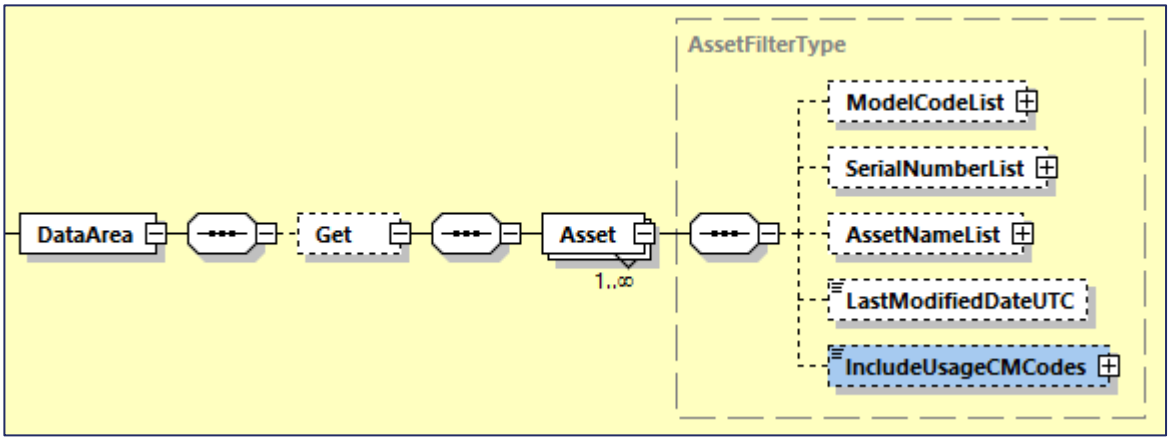


Figure 4 GetAsset DataArea

8.4 ProcessFunctionalLocation

This interface is used to add new functional locations to AMT or edit existing ones.

8.4.1 Business Rules

Field Name	Business Rules	Optional/ Required
FunctionalLocationID	This field is always used to uniquely identify a functional location within AMT. This field must be unique.	R
FunctionalLocationName	This field is always used to uniquely identify a functional location within AMT. This field must be unique.	R
ParentFunctionalLocationID	The parentID must exist in AMT. This field is used in conjunction with the ParentFunctionalLocationName to match an existing FunctionalLocation.	O
ParentFunctionalLocationName	The parentName must exist in AMT. This field is used in conjunction with the ParentFunctionalLocationID to match an existing FunctionalLocation.	O
ModelCode	This value must exist in AMT. The value provided should match a configured **Model** code within AMT. Depending on the AMT configuration, this field is used together with Serial Number and AssetRegistrationCounter to uniquely identify the part within AMT.	O
SerialNumber	This value must exist in AMT. Depending on the AMT configuration, this field can be used together with ModelCode and AssetRegistrationCounter to uniquely identify the part within AMT.	O
RegistrationCounter	Depending on the AMT configuration, this field is used together with the Serial Number and Model Code to uniquely identify the part within AMT.	O

Field Name	Business Rules	Optional/ Required
	If not provided, the currently active registration is used.	
FunctionalLocationTerminatedIndicator	Only the following values are accepted: • **0** – Not Terminated • **1** - Terminated If no value is provided, 0 is used.	O
FunctionalLocationAttribute	The value provided should match a configured **Functional Location Attribute** within AMT.	O

8.4.2 Processing Rules

1. One or more Functional Locations can be included in the same message.
2. Each Functional Location is processed individually. Failure of a Functional Location load is reported, and processing continues.
3. If the Functional Location does not exist:
   a. The Functional Location is created.
   b. Any value against a provided optional tag is used.
   c. If an optional tag is not provided, the value is not set in AMT, unless a specific business rule exists to manage this case.
4. If the Functional Location does exist
   a. The Functional Location will be updated with new provided values.
   b. Any value against a provided tag will be overwritten.
   c. If no tag is provided, the existing values will remain as-is.
5. If the ParentFunctionalLocation does not exist, the Functional Location is created, but a warning is returned. A subsequent process can update the ParentFunctionalLocationID.
6. Functional Location can be duplicated - the Functional Location processed last will take precedence.
7. **RespondFunctionalLocation** is returned upon completion of processing.

8.4.3 Data Structure

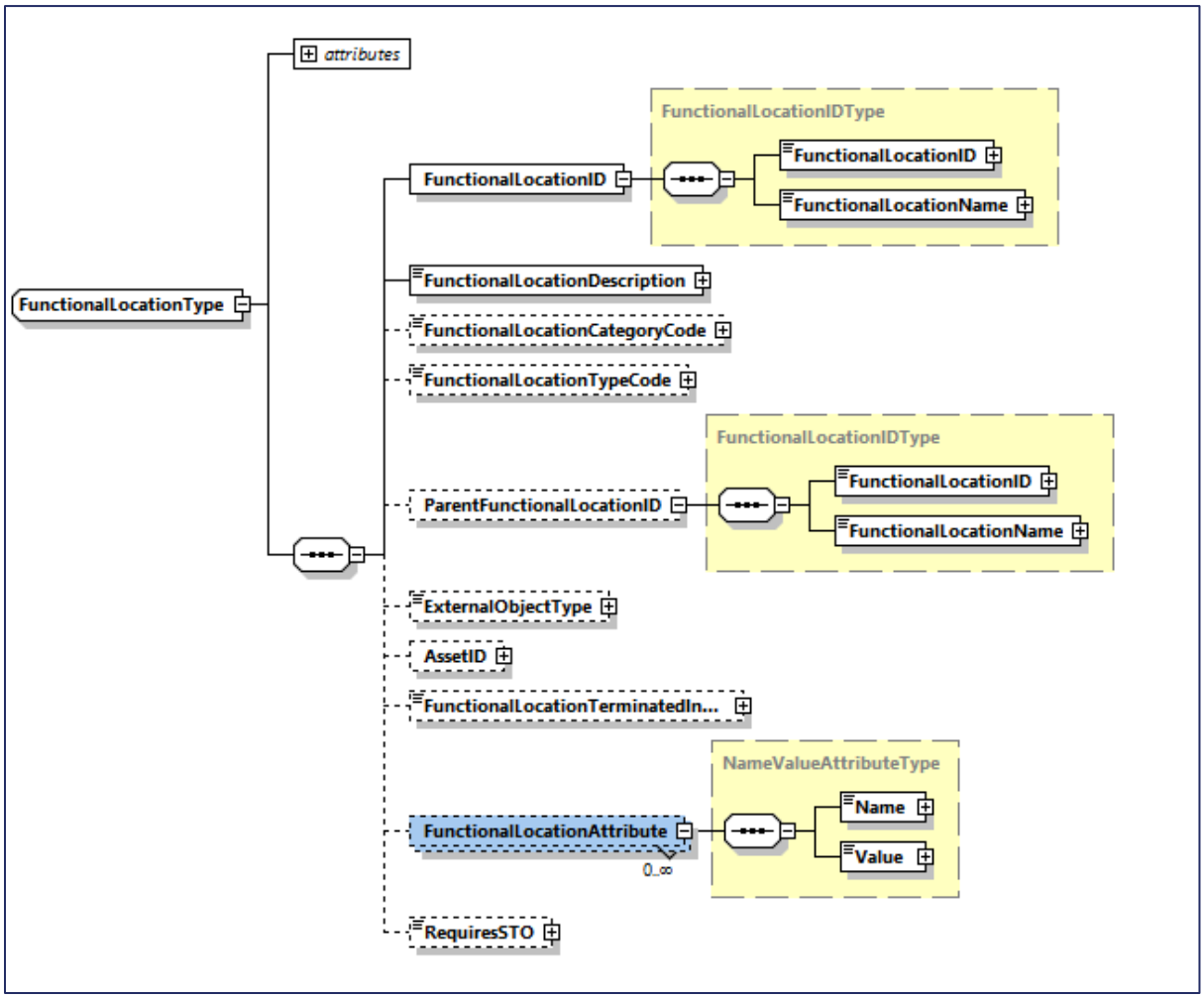


Figure 5 FunctionalLocation

8.5 ProcessFunctionalLocationRelationship

This interface is used to update the functional location hierarchy.

Field Name	Business Rules	Optional/ Required
FunctionalLocationID	This field is always used to uniquely identify a functional location within AMT. This field must be unique.	R
FunctionalLocationName	This field is always used to uniquely identify a functional location within AMT. This field must be unique.	R
ParentFunctionalLocationID	The parentID must exist in AMT. This field is used in conjunction with the ParentFunctionalLocationName to match an existing FunctionalLocation.	R
ParentFunctionalLocationName	The parentName must exist in AMT. This field is used in conjunction with the ParentFunctionalLocationID to match an existing FunctionalLocation.	R

8.5.1 Processing Rules

1. One or more Functional Location Relationships can be included in the same message.
2. Each Functional Location Relationship is processed individually. Failure of a Functional Location Relationship load is reported, and processing continues.
3. If the Functional Location Relationship does not exist, the Functional Location is created.
4. If the Functional Location Relationship does exist, no action is taken.
5. Functional Location can be duplicated - the Functional Location processed last will take precedence.

8.5.2 Data Structure

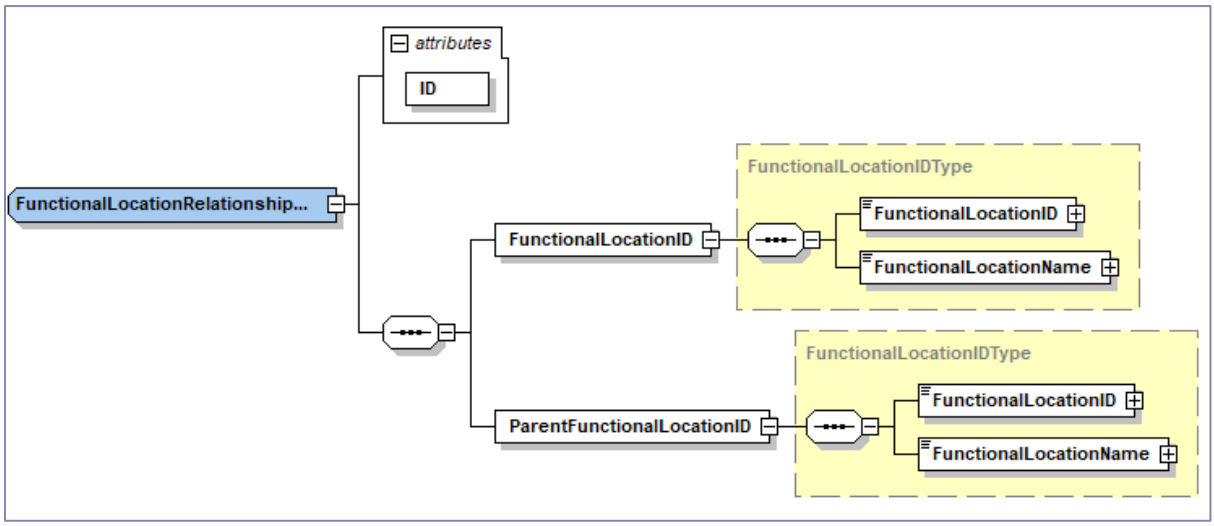


Figure 6 FunctionalLocationRelationship

8.6 CancelFunctionalLocation

This interface marks the functional location as terminated and unlinks it from all assets, strategy tasks and standard jobs.

8.6.1 Business Rules

Field Name	Business Rules	Optional/ Required
FunctionalLocationID	This field is always used to uniquely identify a functional location ID within AMT.	R
FunctionalLocationName	This field is always used to uniquely identify a functional location Name within AMT.	R

8.6.2 Processing Rules

1. One or more Functional Locations can be included in the same message.
2. Each Functional Location is cancelled individually. Failure of a single Functional Location is reported, and processing continues.
3. Functional Locations can be duplicated within the message.
4. If the Functional Location does exist, the **Asset Terminated** flag is set to **True**.
5. If the Functional Location does not exist in AMT, no further action is taken.
6. **RespondFunctionalLocation** is returned upon completion of the process.

8.6.3 Data Structure

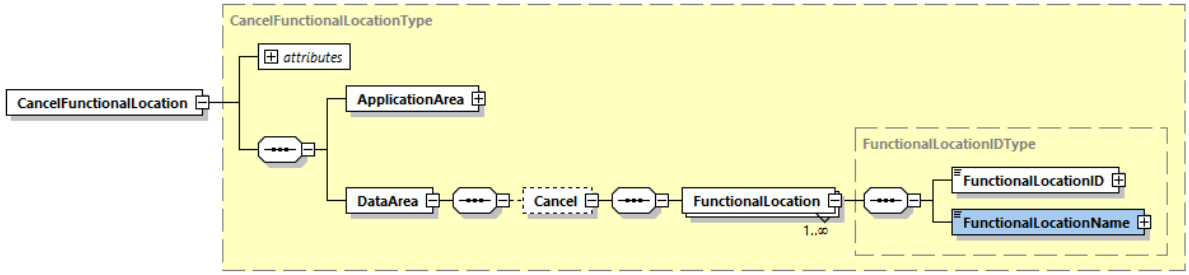


Figure 7 CancelFunctionalLocation DataArea

8.7 GetFunctionalLocation

This interface is used to retrieve a list of assets from AMT.

8.7.1 Business Rules

Field Name	Business Rules	Optional/ Required
FunctionalLocationID	The complete FunctionalLocation unique identifier is matched.	O
FunctionalLocationName	The complete FunctionalLocation unique identifier is matched.	O

8.7.2 Processing Rules

1. Any provided fields will be matched with the existing FunctionalLocation filter in AMT.
2. All items within a list element are considered as a logical OR

   Each list is considered as a logical AND.
3. **ShowFunctionalLocation** is returned upon completion of the process.

8.7.3 Data Structure

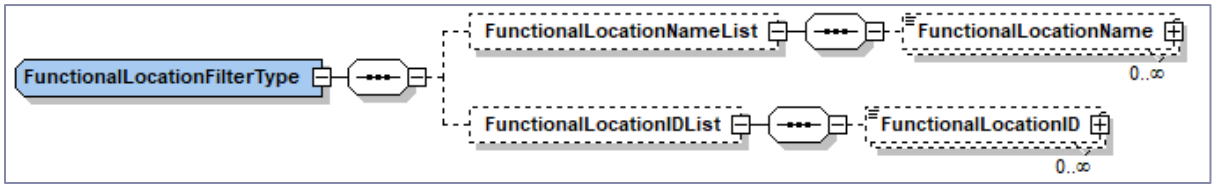


Figure 8 GetFunctionalLocation DataArea

9 Response Formats

9.1 RespondAsset

This payload is returned in response to the following interfaces:

- ProcessAsset
- CancelAsset

9.1.1 Data Structure

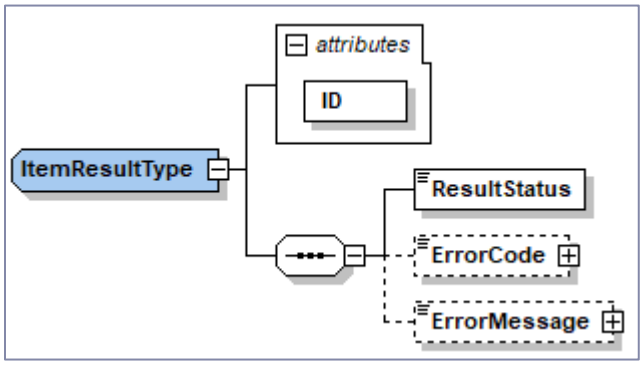


Figure 9 RespondAsset

9.2 ShowAsset

This payload is returned in response to the following interfaces:

- GetAsset

9.2.1 Data Structure

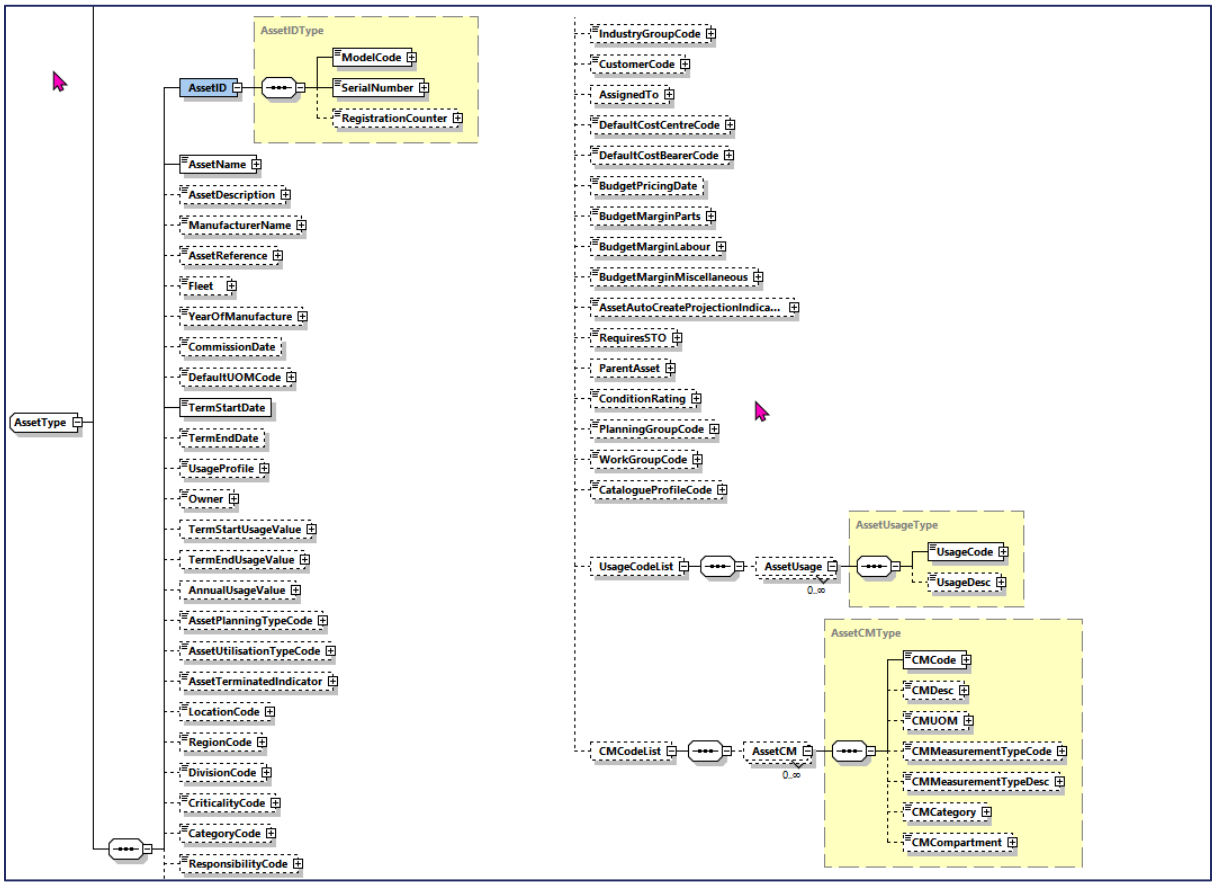


Figure 10 ShowAsset

9.3 RespondFunctionalLocation

This payload is returned in response to the following interfaces:

- ProcessFunctionalLocation
- CancelFunctionalLocation
- ProcessFunctionalLocationRelationship

9.3.1 Data Structure

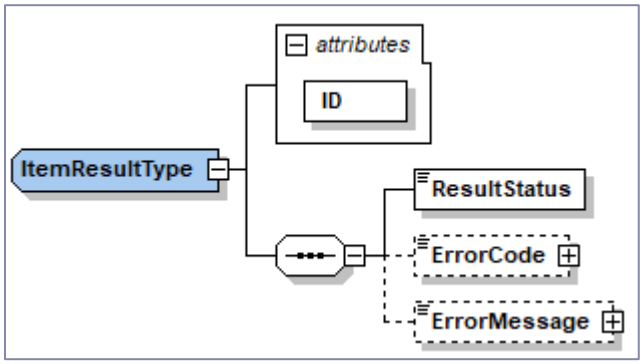


Figure 11 RespondFunctionalLocation

9.4 ShowFunctionalLocation

This payload is returned in response to the following interfaces:

- GetFunctionalLocation

9.4.1 Data Structure

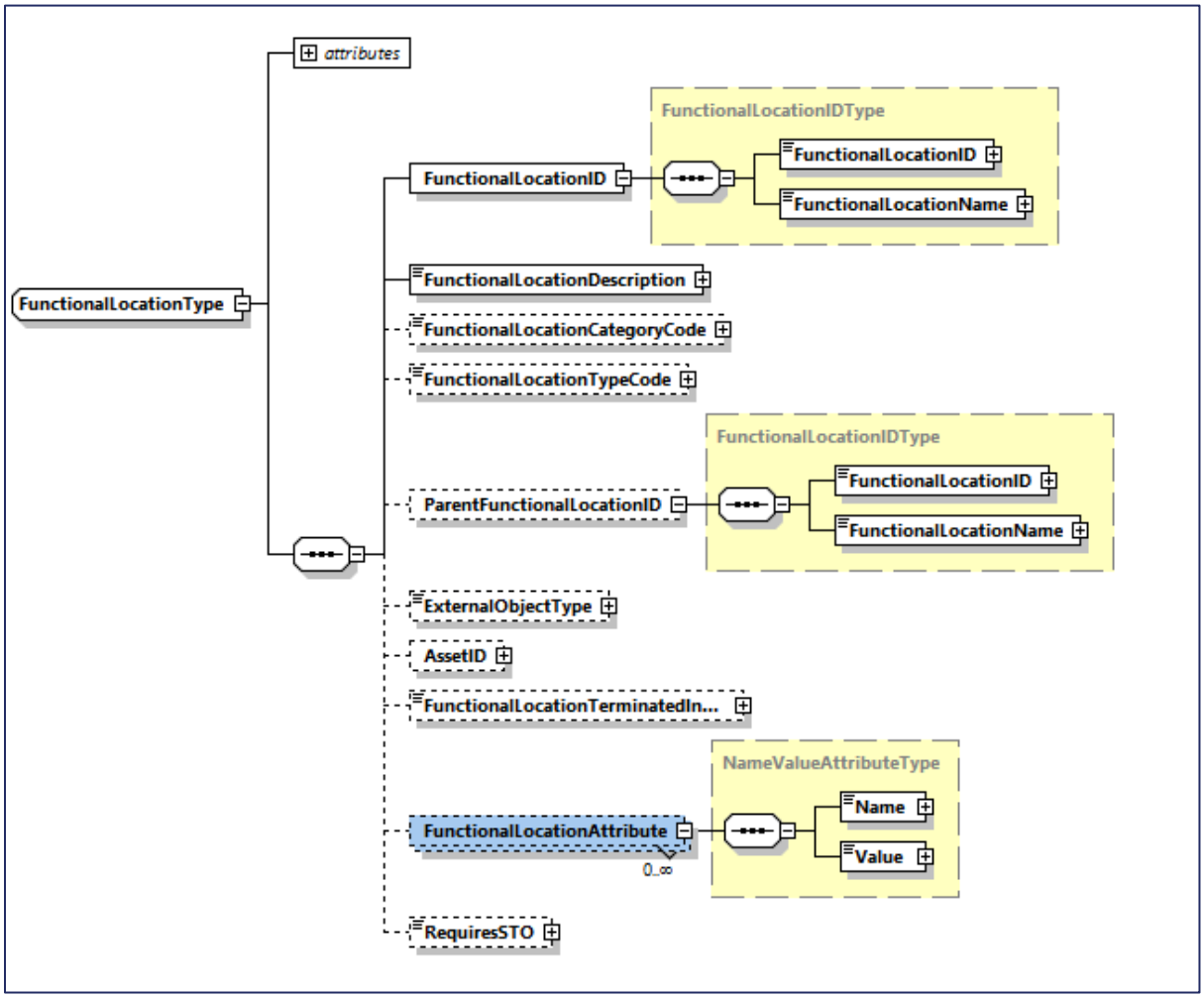


Figure 12 ShowFunctionalLocation